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Anti-inflammatory and anti-nociceptive activities of *Alpinia Oxyphylla* Miquel extracts in animal models

Su Hyun Yu^a, Hyun Jin Kim^a, Se Yeong Jeon^a, Mi Ran Kim^a, Bo Su Lee^a, Jeong Jun Lee^a, Dong-Seon Kim^b, Young Chul Lee^{a,*}

^a R&D Center, Naturetech Co., Ltd., 450-86, Maeboong-Ro, Dongnam-Gu, Cheonan-Si, Chungnam, 31257, South Korea

^b Herbal Medicine Research Division, Korea Institute of Oriental Medicine, 1672, Yuseong-Daero, Yuseong-Gu, Daejeon, 34054, South Korea

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ABSTRACT

Ethnopharmacological relevance: *Alpinia Oxyphylla* Miquel is a widely used traditional herbal medicine for the treatment of abdominal pain, intestinal disorders, enuresis, and inflammatory conditions.

Aim of the study: In this study we aimed to demonstrate the anti-inflammatory and anti-nociceptive effects of *Alpinia Oxyphylla* Miquel extracts with 50% ethanol (AOE) in animal models.

Material and method: The anti-inflammatory and nociceptive effects of AOE were estimated *in vivo* using a carrageenan-induced rat paw edema model, an arachidonic acid-induced mice ear edema model, an acetic acid induced mice writhing test model, and an *ex vivo* sulfated glycosaminoglycan (GAG) inhibition assay. *In vitro* inhibition assays for cyclooxygenase (COX) and lipoxygenase (LOX) were performed to determine the IC₅₀ of AOE.

Results: Compared with the control, AOE caused a significant inhibition of paw edema, ear thickness, and abdominal constriction in animals. Compared with untreated explants exposed to interleukin-1 alpha (IL-1α), proteoglycan degradation by IL-1α was inhibited (18.2%–82.0%) in explants treated with AOE (50–400 μg/ml).

In vitro, IC₅₀ values of AOE were 299.87, 199.07, and 170.29 μg/ml for COX-1, COX-2, and 5-LOX respectively.

Conclusions: These data demonstrate the potent anti-inflammatory and analgesic effects of AOE and suggest AOE to be useful for the treatment of osteoarthritis and/or associated symptoms.

1. Introduction

Alpinia oxyphylla Miquel (*A. oxyphylla*) grows in the tropical and subtropical regions China (Zhao et al., 2016). It has been used food and folk medicines for centuries and called Ikjin and Yizhi Ren in Korea and China. According to the Chinese Pharmacopoeia (But et al., 1997; Pharmacopoeia, 2005), one of the famous south medicines in China is widely used as an anti-diarrheal, treatment of abdominal pain, and anti-polyuric. *A. oxyphylla* is used as an anti-polyuric according to the Donguibogam (Principles and Practice of Eastern medicine) (Heo, 1994) in Korea. *A. oxyphylla* exhibit ethnopharmacological activities via, novel phytochemicals and bioactive compounds such as sesquiterpenes, flavonoids, diarylheptanoids, and volatile oils (Itokawa et al., 1982, 1981; Muraoka et al., 1998). *A. oxyphylla* has numerous biochemical activities including anti-tumor (Chun et al., 2002b; Lee et al., 1998), anti-anaphylactic (Shin et al., 2001), anti-ulcer (Yamahara et al.,

1990), neuroprotective activities (An et al., 2006; Yu et al., 2003), anti-oxidant (Han et al., 2007), anti-inflammatory (Chun et al., 2002b), and anti-angiogenic effects (He et al., 2010). Among numerous biochemical activities of *A. oxyphylla*, the focus was on the study of anti-inflammatory effects. For example, the study using lipopolysaccharides (LPS) and interferon-gamma (IFN-γ) stimulated RAW 264.7 cell reported that the anti-inflammatory effects of *A. oxyphylla* occur owing to the inhibition of nitric oxide (NO) production, prostaglandin E2 release, and pro-inflammatory cytokines response (Xu et al., 2012). In addition, nootkatone (sesquiterpenes) derived from *A. oxyphylla* exert anti-inflammatory effects on PBL-2H3 cells (mouse peritoneal macrophages) (Morikawa et al., 2002). Yakuchinone A and B (diarylheptanoids) suppress tetradecanoylphorbol (TPA)-stimulated production of COX-2 and, tumor necrosis factor-alpha (TNF-α) in mouse ear skin (Jiang et al., 2013). Recently a benzylidenacetophenone derivative was synthesized from yakuchinone B it exerted anti-arthritis effect and

* Corresponding author.

E-mail addresses: yuhut@naturetech.co.kr (S.H. Yu), hjmari@naturetech.co.kr (H.J. Kim), junsy1220@naturetech.co.kr (S.Y. Jeon), mrkim@naturetech.co.kr (M.R. Kim), bslee@naturetech.co.kr (B.S. Lee), jjlee@naturetech.co.kr (J.J. Lee), dskim@kiom.re.kr (D.-S. Kim), ychee@naturetech.co.kr (Y.C. Lee).

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inhibited pro-inflammatory cytokines in carrageenan/kaoline-induced rat model (Chun et al., 2002b, 2002a). Although these studies highlight the anti-inflammatory potential of *A. oxyphylla* and its bioactive compounds, little is known regarding the anti-nociceptive activity of *A. oxyphylla* in animal models. Therefore, we examined *A. oxyphylla* as a potential food supplement for treating inflammatory arthritis based on the anti-inflammatory effects of *A. oxyphylla*.

Inflammatory responses are important host defense mechanism and often include symptoms of redness, swelling, pain, and fever. Hosts respond to external stimuli or tissue damage via increased production of pro-inflammatory cytokines such as TNF- α , interleukin-1 beta (IL-1 β), interleukin-6 (IL-6), NO and prostaglandins (PGs) (Goldring and Otero, 2014). TNF- α and IL-1 β are the major pro-inflammatory mediators involved in inflammation and pain during tissue damage caused by various stimuli. In addition, NO and PGs mediate inflammation and pain, which are produced by inducible NO synthase (iNOS) and COX-2, respectively (Süleyman and Büyükköroğlu, 2001). COX-1, COX-2, and LOX convert arachidonic acid to PGs, which stimulates inflammation (Bouclier et al., 1990). COX-2 is not expressed in healthy tissue; its expression is tightly regulated and only induced in response to cellular stress or damage via the activation of growth factor pathways. Inflammation also alters arachidonic acid metabolism via LOX activity which signals for the synthesis of pro-inflammatory cytokines such as IL-1, IL-6, and interleukin-8 (IL-8) (Kalaiselvi et al., 2013).

Currently used arthritis drugs for pain management such as steroids and non-steroidal anti-inflammatory drugs (NSAIDs) are believed to function, in part, by suppressing COX-mediated PG formation (Wolfe et al., 1999; Sanghi et al., 2012). While effective for pain management, steroids and NSAIDs do not correct the actual etiology of arthritis, which leads to irreversible joint damage. When taken long term steroids and NSAIDs also have serious adverse gastrointestinal (GI) and cardiovascular side effects (Flower, 1974). Our previous study demonstrated that, AOE inhibit pro-inflammatory cytokines and cartilage degradation using monosodium iodoacetate-induced rat osteoarthritis model (Lee et al., 2019). In this study aimed, we demonstrate that the traditional medicine herb AOE may be an excellent alternative to steroids and NSAIDs for the treatment of arthritis and other inflammatory diseases.

2. Materials and methods

2.1. Materials

In this study, we used AOE containing 5.81 ± 0.21 mg/g of nootkatone, which is the major compound of this AOE as demonstrated by HPLC analysis in a previous study (Lee et al., 2019). A voucher specimen (2-17-0580) was deposited at Korean herbarium of standard Herbal Resources of Korea Institute of Oriental Medicine (Daejeon, Korea). *A. oxyphylla* was extracted under reflux with 50% ethanol twice for 4 h. The extract was filtered, concentrated and dried.

2.2. Animal

Sprague-Dawley rats (CrI:CD, 6-week old) for carrageenan-induced paw edema assay, ICR mice (CrI:Ori:CD-1, 6-week old) for arachidonic acid-induced ear edema, and acetic acid-induced writhing test were obtained from OrientBio Co., Ltd. (Seongnam, Korea). Rats and mice were acclimated to the laboratory for a week and housed in specific pathogen room-free rooms ($23.0 \pm 1.0^\circ\text{C}$ with humidity $50 \pm 10\%$ under a 12 h light/dark cycle). Drugs were delivered as individual. Doses to animals based on their measured bodyweight and were adjusted to maintain the targeted dose level for all rats and mice. New Zealand white rabbits (Yac:NZW, 12-week old, 2.0 kg) were also purchased from OrientBio for GAG release inhibition assay. All studies were approved by the Institutional Animal Care and Use Committees of Naturetech, Inc. based on Animal Protection Act (Approval No. NT21803, NT21901, NT21903, and NT21904).

2.3. COX-LOX assay

We employed the commercial colorimetric COX (ovine) inhibitor screening assay kit (Cayman Chem., Co., Cat 760111) according to the manufacturer's protocol. LOX inhibition measured using the commercial 5-LOX (potato, Cat#60401) inhibitor screening assay kit (Cayman Chem., Co., Cat760700) according to the manufacturer's protocol. AOE (50, 100, 200, 400 $\mu\text{g/ml}$) was assayed for COX and LOX reaction. The AOE concentration that resulted in 50% inhibition of COX and LOX release (IC_{50}) was calculated from the concentration inhibition value, respectively.

2.4. Carrageenan induced paw edema

Carrageenan-induced rat hind paw edema, an evaluation model of acute inflammatory response, was used with some modification (Yimam et al., 2016). Sprague-Dawley rats were adapted to the treatment room for 30 min and baseline values of the hind paw were measured using a plethysmometer (2390 series Electrovonfrey, IITC, Woodland Hills, CA) before carrageenan injection. Control groups were administered with 0.5% carboxymethyl cellulose (CMC) and indomethacin (in 0.5% CMC) orally at 10 mg/kg/body weight. Experimental groups were treated with two doses of AOE in 0.5% CMC (in saline) at either 100 or, 300 mg/kg/body weight before injection of carrageenan. Topical inflammation was induced by intraplantar injection of carrageenan λ (Sigma Aldrich, St Louis) into the surface of right hind paw of rat under anesthesia with 2% isoflurane. Paw edema was measured at before injection of carrageenan, and at 1, 3, and 5 h post injection. Edema was measured by paw volume ($V_{\text{time}} - V_{\text{zero}}$) over time. V_{zero} is the baseline of paw volume for each group before carrageenan injection. V_{time} is the paw volume for each group at time at 1, 3, and 5 h after carrageenan injection. The percentage of paw edema was calculated by the following equation.

$$\text{Percentage of paw edema} = \frac{(V_{\text{time}} - V_{\text{zero}})}{V_{\text{zero}}} \times 100$$

2.5. Arachidonic acid induced ear edema

Arachidonic acid-induced mouse ear edema is, an acute nociceptive effect model and, was used with some modification (Yimam et al., 2015). The ICR mice were divided into five groups consisting of arachidonic acid induced control, indomethacin-treated (5 mg/kg/body weight), and AOE-treated (150, 300, 600 mg/kg/body weight). Mice were orally administered AOE 150, 300, 600 mg/kg twice (24 h and 1 h before application of arachidonic acid). Measurements of ear thickness were taken using a micrometer clipper before arachidonic acid (suspended in acetone) was applied. One hour after the last treatment, arachidonic acid was applied topically at a dose of 2 mg/20 μl /ear to the inner and outer surface of the right ear. Thirty minutes after induction, ear swelling measurements were taken again. After measurements of thickness, ear tissue was harvested and fixed in 10% buffered formalin. The tissues for histological analysis were stained by hematoxylin and eosin (H&E). The percent change of ear thickness was calculated by the ear thickness ($T_{\text{time}} - T_{\text{zero}}$). The percentage of ear swelling thickness was calculated as per the carrageenan-induced paw edema model.

2.6. Acetic acid induced writhing response

Acetic acid-induced mice abdominal writhing is, an evaluation model for nociceptive effects and, was used with some modification (Yimam et al., 2012). ICR mice were divided into five groups consisting of acetic acid induced control, an indomethacin treatment group (5 mg/kg/bodyweight), and three AOE treatment groups (150, 300, 600 mg/kg/bodyweight). Mice were orally administered AOE twice before

acetic acid (0.7% in saline). Thirty minutes after the last treatment, acetic acid was intraperitoneal administrated at 10 ml/kg/bodyweight. After acetic acid treatment, each mouse was placed into solitary observation chamber. Following the abdominal muscle constrictions were measured over 30 min.

2.7. Ex vivo GAG release inhibition assay

Rabbit articular cartilages (2.0 kg bodyweight) from hock joints were harvested immediately after animal sacrifice. Articular cartilage explants were acquired as previously described (Cho et al., 2006; Yimam et al., 2012). After surgically exposing the articular surfaces of the femur and tibia under sterile conditions, approximately 200–220 mg of surface per joint were dissected and soaked in complete medium (Dulbecco's modified Eagle's medium, 5% heat inactivated fetal bovine serum, penicillin 100 µg/ml, streptomycin 100 µg/ml) for 2 days. - Following, 30 mg cartilage pieces were placed in 48-well plates and treated with AOE (50, 100, 200, 400 µg/ml) and diclofenac (200 µg/ml) diluted with a basal medium (Dulbecco's modified Eagle's medium, 1% heat inactivated fetal bovine serum, penicillin 100 µg/ml, streptomycin 100 µg/ml, HEPES 10 mM). -After 1 h, 5 ng/ml of recombinant human interleukin-1 alpha (rhIL-1α) was added to the medium and incubated at 37 °C in a humidified 5% CO₂ for 24 h. Sulfated GAG concentration was determined by using a commercially available kit (Glycosaminoglycan assay Blyscan, Biocolor, B1000) according to the manufacturer's protocol.

2.8. Statistical analysis

The statistical analyses were conducted using the SAS (version 9.2 SAS institute Inc., Cary, NC). All data obtained from paw edema, ear swelling, and abdominal contraction experiments were analyzed using the Analysis of Variance followed by Tukey's multiple comparison test. The results are presented as the mean and standard deviation. A value of $P < 0.05$ was considered significant.

3. Result

3.1. COX and LOX inhibition

AOE suppressed COX-1 and COX-2 enzyme activity in a dose dependent manner with IC₅₀ 299.89 µg/ml and 199.07 µg/ml, respectively. In addition, 5-LOX was also inhibited with an IC₅₀ 170.29 µg/ml.

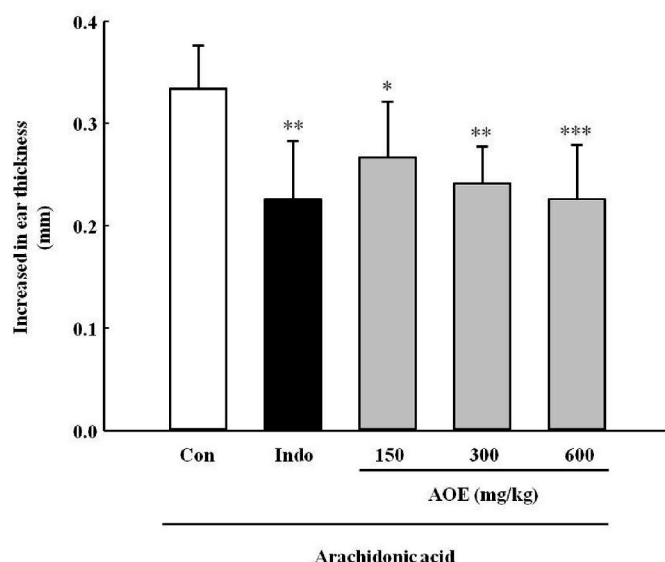
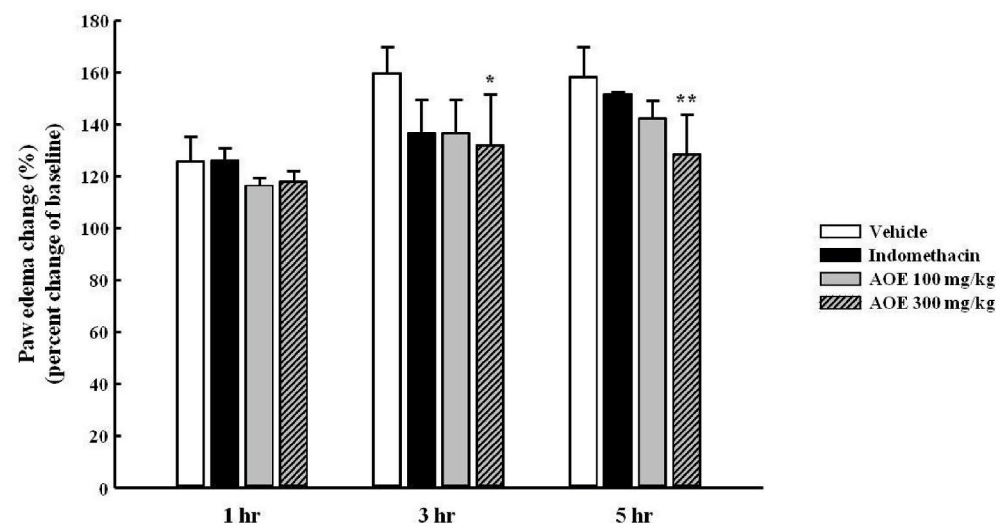


Fig. 2. Inhibitory effect of AOE 150, 300, 600 mg/kg, and indomethacin 5 mg/kg on ear thickness changes as measured by microclipper on arachidonic acid-induced mice. The increased ear thickness was evaluated at 30 min after arachidonic acid application. Results are mean $\pm$ SD (standard deviations). * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$ indicate statistically significant differences from control.

3.2. Effect of AOE on carrageenan-induced paw edema

As shown in Fig. 1, edema reached a maximum (percentage change of baseline) 3 h after the injection. Significant differences between control and treated rats were still observed 5 h after treatment. Rat pretreated with AOE (300 mg/kg) had significantly less paw edema (32.8% 3 h, 32.4% 5 h) when compared with the vehicle control group.

3.3. Effect of AOE on arachidonic acid induced ear swelling

As shown in Fig. 2, topical application of arachidonic acid-induced an acute inflammatory response. Micro-clipper measurements demonstrate that ear swelling of AOE-treated mice (150, 300, and 600 mg/kg) is decreased to 0.267 ± 0.054 mm, 0.241 ± 0.036 mm, and 0.226 ± 0.053 mm respectively. These results are statistically significantly when compared with control (0.334 ± 0.042 mm). AOE reduced swelling similar to the indomethacin, positive control group (0.226 ± 0.057 mm), making it similarly effective. H&E-stained ear

Fig. 1. Inhibitory effects of AOE 100, 300 mg/kg, and indomethacin 10 mg/kg on changes in the right hind paw edema in carrageenan-induced rats. The volumes of paw edema were evaluated at 1, 3, and 5 h after carrageenan injection. Results are mean $\pm$ SD (standard deviations). * $P < 0.05$ and ** $P < 0.01$ denote statistically significant results.

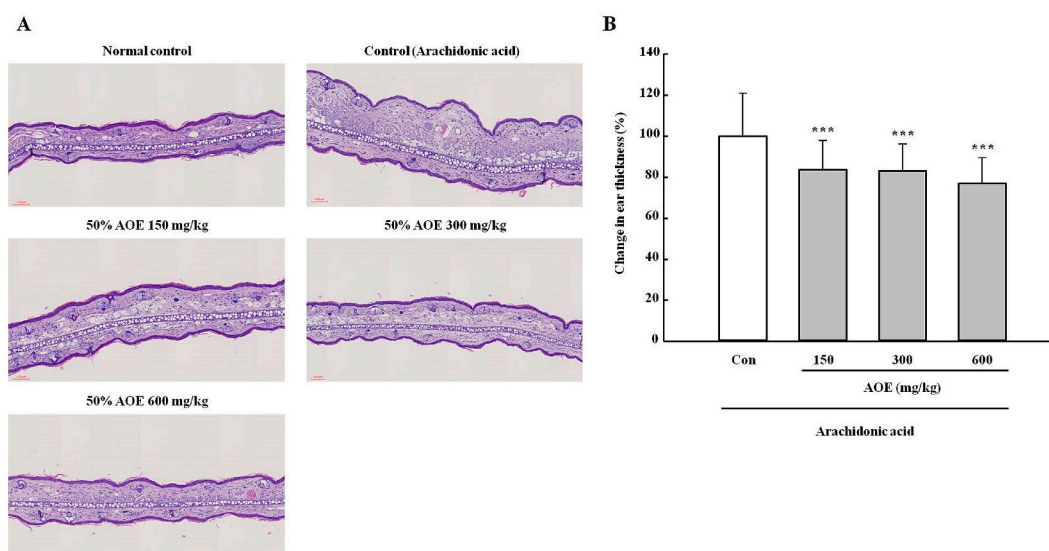


Fig. 3. Inhibitory effect of AOE 150, 300, 600 mg/kg on mouse ear thickness in arachidonic acid-induced edema. (A) Ear tissue after arachidonic acid application was H&E stained. Scale bar 100 μ m. (B) Changes percentage in thickness of ear tissue stained H&E. Results are mean $\pm$ SD (standard deviations). *** P < 0.001 indicate statistically significant differences from control.

tissue sections were used to evaluate histological and changes percentage of ear swelling thickness (Fig. 3A and B). The changes percentage in ear thickness were significantly lower in AOE treatment group at 150, 300, and 600 mg/kg (83.5%, 83.0%, and 77.1%) when compared to control.

3.4. Effect of AOE on acetic acid induced writhing response

As observed in Fig. 4, AOE treatments (150, 300, and 600 mg/kg) and indomethacin (5 mg/kg) decreased abdominal constrictions with acetic acid injection when compared to control. Oral administration of 50% AOE at 300, 600 mg/kg and indomethacin 5 mg/kg significantly prevented acetic acid induced abdominal writhing with inhibition of 53.3%, 52.9%, and 88.4% respectively.

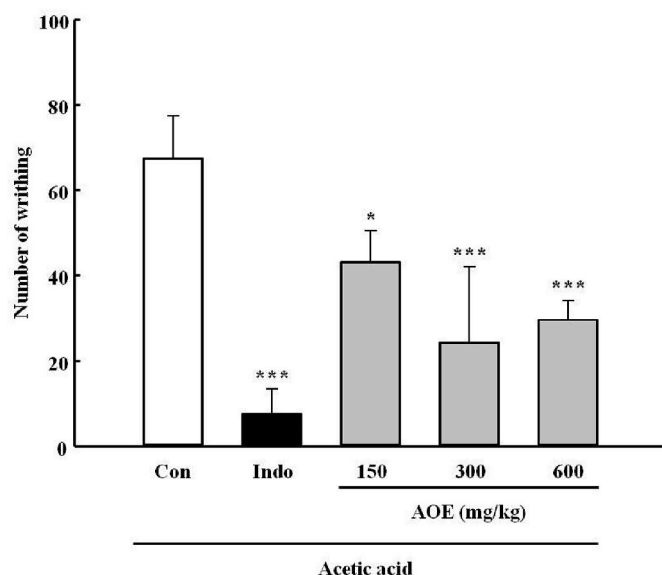


Fig. 4. Inhibitory effect of AOE 150, 300, 600 mg/kg on the changes in mouse abdominal constriction when treated with acetic acid. Abdominal writhing was evaluated for 30 min post treatment. Results are mean $\pm$ SD (standard deviations). * P < 0.05 and *** P < 0.001 indicate statistically significant differences from control.

3.5. GAG release inhibition

The AOE (50, 100, 200, 400 μ g/ml) and diclofenac (200 μ g/ml) inhibited the release of GAG from IL-1 α treated rabbit hock joints ($85.0 \pm 5.0\%$, $70.5 \pm 1.8\%$, $60.5 \pm 2.7\%$, $32.3 \pm 2.6\%$, and $17.3 \pm 2.9\%$ respectively) in a concentration dependent manner of AOE (Fig. 5). In addition, in the rhIL-1 α -treated group, the percentage of GAG released was significantly reduced when compared to control. AOE at 400 μ g/ml nearly ameliorates the rhIL-1 α -induction of GAG.

4. Discussion

The present study demonstrates the anti-inflammatory and analgesic effects of AOE in multiple model systems including carrageenan-induced rat paw edema, arachidonic acid-induced mice ear edema, and acetic acid-induced mice writhing response. Further evidence for the anti-inflammatory effects of AOE are provided by our *ex vivo* in GAG

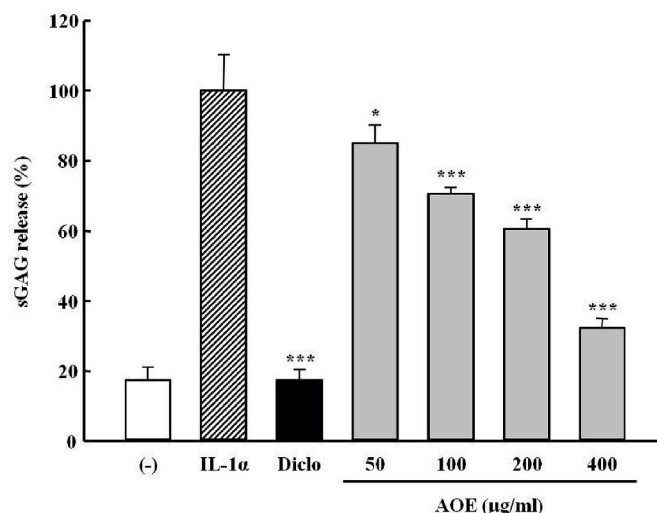


Fig. 5. Inhibitory effect of AOE 50, 100, 200, 400 μ g/ml on the release of GAG from rhIL-1 α induced rabbit cartilage explants. Results are mean $\pm$ SD (standard deviations). * P < 0.05 and *** P < 0.001 indicate statistically significant differences from pretreated control.

assays, which also suggest that AOE could have protective effects against articular cartilage degradation.

In general, NSAIDs such as aspirin are the most widely used prescription drugs for the treatment of degenerative arthritis, rheumatoid arthritis, and headaches with anti-inflammatory, antipyretic and analgesic effects (Liao et al., 2014). However, when NSAIDs are used for osteoarthritis, the symptoms may improve, but the loss of joint cartilage and disease progression cannot be prevented. In addition, Wallace, and colleagues demonstrated that half of the patients taking NSAIDs showed serious side effects such as gastrointestinal dysfunction and stopped taking it within a year (Wallace et al., 1997). The main mechanism of NSAIDs is believed to be COX inhibition, although this has yet to be definitively demonstrated. Recently, a COX-2 “selective” inhibitor with reduced GI side effects was demonstrated to suppress COX-2 and 5-LOX. However, this “selective” inhibitor has other side effects such as cardiovascular disease due to imbalances in the COX pathway (Ju et al., 2019).

LOX is another enzyme involved in the process by which arachidonic acid is metabolized by chemical mediators in inflammatory reactions (Bogdan, 2001). There are two known isoform types of COX: COX-1 and COX-2 (Simon, 1999). COX-1 is persistently expressed in the most tissue and play a house-keeping role for the regulation of PGs under normal physiological conditions such as gastric mucosa maintenance, kidney function, and platelet regulation (Honda et al., 2010). COX-2 is normally expressed in response to growth factors that are released under both disease and physiological conditions. In particular, COX-2 is induced by cytokines that cause pro-inflammation. To date, many NSAIDs used to treat inflammation also inhibit COX-1; thus, the appearance side effects such as gastric mucosal corrosion and ulcers are common (Kabashima and Miyachi, 2004). Given the failure of pharmaceutical COX inhibitors to treat arthritis, natural inhibitors of COX enzymes may be a new therapy for the inflammatory disease and are being sought after.

COX catalyze arachidonic acid to PG (Ferreira, 1981). PGs are a central mediator of inflammation responses and pain was shown by many studies (Ricciotti and FitzGerald, 2011; Sulaiman et al., 2008). PGs can also stimulate smooth muscle contraction, and in some animals decrease or increase blood pressure and aggregation. In addition, PGs can promote membrane ion transport, inflammation, and prevent lipolysis in adipose tissue.

Our results demonstrate that, the IC_{50} for AOE inhibition of COX-1, COX-2, and 5-LOX are all in a similar range (100–300 μ M), consistent with the suppression of inflammation and the analgesic effects we observe in all of our model systems. For example, AOE treatment (300 mg/kg) significant reduced carrageenan-induced rat paw edema (Fig. 1). In anti-inflammatory and anti-nociceptive studies, carrageenan-induced paw edema model has been accepted as useful tools, characterized by cellular and molecular mechanism (Eddouks et al., 2012). Given that indomethacin and NSAIDs inhibit COX, we propose that AOE works through a similar mechanism to exert its inflammatory and analgesic effects. The anti-inflammatory activity of AOE was also confirmed in our mouse ear inflammation model using arachidonic acid (Figs. 2 and 3). Arachidonic acid rapidly induces edema and is associated with increased LOX and COX products (Rao et al., 1993). Our results indicate that AOE significantly reduces arachidonic acid edema by as much as 23% (Fig. 3). In our acetic acid abdominal constriction test (Fig. 4), acetic acid indirectly acts on the release of mediators such as bradykinin, serotonin, histamine, pro-inflammatory cytokines (TNF- α , IL-1 β , IL-8) and PGs. AOE treatment has significant anti-nociceptive effects in mice treated with acetic acid, reducing abdominal constrictions by as much as 64% (Fig. 4). As GAG excretion is a clinical symptom of osteoarthritis (Kim et al., 2012), our results in Fig. 5 demonstrating the reduction of GAG release in mice treated with AOE, strongly suggest that AOE could be an effective treatment. When compared to the effective yet non-selective COX-2 inhibitor diclofenac, AOE treatment (400 μ g/ml) had a similar inhibitory effect on GAG

release, which is coherent the results from other studies on cartilage metabolism. Overall, our results suggest that AOE exerts its effect by interfering with inflammatory signaling pathways. Importantly, the anti-inflammatory and analgesic effects of AOE in our study appear to be at the same concentrations used in previous studies (Lee et al., 2019). Further studies are needed to determine whether cartilage regeneration and/or protection by AOE occurs in addition to its analgesic and anti-inflammatory effects.

5. Conclusions

The data presented here clearly demonstrates the anti-inflammatory and anti-nociceptive effects of AOE *in vitro* and in animal models of rat carrageenan induced paw edema, mice arachidonic acid induced ear edema and mice acetic acid induced abdominal constriction. The mechanism for AOE's anti-inflammatory and anti-nociceptive effect and potential cartilage regeneration are still unclear; thus, further research is needed.

Declaration of competing interest

The author declare that there are no conflicts of interest.

Acknowledgment

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Abbreviations

A. oxyphylla	Alpinia Oxyphylla Miquel
AOE	Alpinia Oxyphylla Miquel extracts
COX	cyclooxygenase
LOX	lipoyxygenase
IL-1 α	interleukin-1 alpha
IFN- γ	interferon-gamma
NO	nitric oxide
LPS	lipopolysaccharides
TPA	tetradecanoylphorbol acetate
TNF- α	tumor necrosis factor-alpha
IL-1 β	interleukin-1 beta
IL-6	interleukin-6
IL-8	interleukin-8
PGs	prostaglandins
iNOS	inducible nitric oxide synthase
NSAIDs	non-steroidal anti-inflammatory drugs
GI	gastrointestinal
CMC	carboxymethyl cellulose
H&E	hematoxylin and eosin
GAG	glycosaminoglycan
rhIL-1 α	recombinant human interleukin-1 alpha

CRedit authorship contribution statement

Su Hyun Yu: Methodology, Investigation, Formal analysis, Writing original draft. **Hyun Jin Kim:** Investigation, Data Curation. **Se Yeong Jeon:** Formal analysis, Investigation. **Mi Ran Kim:** Formal analysis, Investigation. **Bo Su Lee:** Investigation, Resource. **Jeong Jun Lee:** Supervision, Project administration. **Dong-Seon Kim:** Supervision, Project administration. **Young Chul Lee:** Conceptualization, Methodology, Data Curation, Writing-reviewing and editing.

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Su Hyun Yu^a, Hyun Jin Kim^a, Se Yeong Jeon^a, Mi Ran Kim^a, Bo Su Lee^a, Jeong Jun Lee^a, Dong-Seon Kim^b, Young Chul Lee^{a,*}

^a R&D Center, Naturetech Co., Ltd., 450-86, Maebong-Ro, Dongnam-Gu, Cheonan-Si, Chungnam, 31257, South Korea

^b Herbal Medicine Research Division, Korea Institute of Oriental Medicine, 1672, Yuseong-Daero, Yuseong-Gu, Daejeon, 34054, South Korea

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* Corresponding author.

E-mail addresses: yuhut@naturetech.co.kr (S.H. Yu), hjmari@naturetech.co.kr (H.J. Kim), junsy1220@naturetech.co.kr (S.Y. Jeon), mrkim@naturetech.co.kr (M.R. Kim), bslee@naturetech.co.kr (B.S. Lee), jilee@naturetech.co.kr (J.J. Lee), dkim@kiom.re.kr (D.-S. Kim), yclee@naturetech.co.kr (Y.C. Lee).

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